Question assigned to the following page: 1

CS771: Introduction to Machine Learning

Minor Assignment 3

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Experimental Setup

Dataset. We use the `DD1(Oct).csv` AQI dataset, retaining only rows where `Valid`=1. After filtering, there are **10,458** valid timestamps.

Features and label. The target variable is `Ref. O3(ppb)`. The five input features are `Time`, `Temp(C)`, `RH(%)`, `o3op1(mV)`, `o3op2(mV)`. The `Time` column is converted to hour-of-day (0–23) by parsing the timestamp string. The features `Ref. NO2(ppb)`, `no2op1(mV)`, `no2op2(mV)` are excluded throughout.

Train/test split. We use the first 80% of valid timestamps as training data (**8,366** samples) and the remaining 20% as test data (**2,092** samples). No random shuffling is applied; chronological order is preserved exactly as in the lecture code.

Pre-processing. All features are standardised (zero mean, unit variance) using statistics computed *only* on the training set; the same transform is applied to the test set. The label `Ref. O3(ppb)` is likewise standardised for training and de-standardised before computing test MAE, so all reported values are in the original ppb units.

EM Mixed Regression. We follow the lecture code (`lec16.ipynb`).

- **E-step**: assign each training point to the component whose Ridge regression model has the smallest absolute residual.
- **M-step**: re-fit each Ridge model (`alpha`=0.01) using the points assigned to it, weighted by the soft responsibility.
- **Initialisation**: training data are sorted by O3 label value and split uniformly into `n_components` buckets; one Ridge model is fitted per bucket. This sorted initialisation ensures that the EM starts with components already roughly separated by O3 level.
- **Auxiliary classifier**: after EM convergence an RBF-kernel SVM (`C`=1, `gamma='scale'`) is trained on the training assignments. At test time it routes each test point to its predicted component's regression model. All reported test MAEs use this classifier (not oracle assignment), making the reported numbers realistic and comparable to the lecture code result of ~9.16 ppb.

1 Part 1 — Effect of Number of EM Iterations

We fix n_components=4 (as in the lecture code) and vary n_iter over $\{1, 2, 3, 4, 6, 8, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$. Figure 1 shows test MAE as a function of n_iter, and Table 1 lists the precise values.

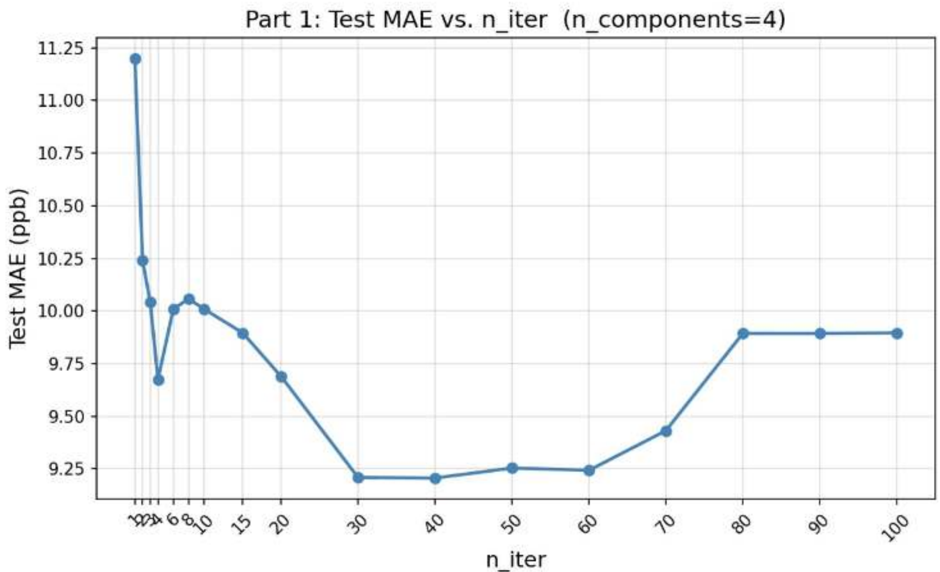

Figure 1: Test MAE vs. n_iter (n_components=4). The curve shows a noisy, generally decreasing trend for small n_iter, stabilises around n_iter=30–40 (the knee), and then *degrades* noticeably beyond iteration 60.

Table 1: Test MAE (ppb) for different values of n_iter (n_components=4).

n_iter	MAE (ppb)	n_iter	MAE (ppb)	n_iter	MAE (ppb)
1	11.24	10	10.01	50	9.25
2	10.24	15	9.90	60	9.24
3	10.04	20	9.69	70	9.43
4	9.67	30	9.21	80	9.89
6	10.01	**40**	**9.21**	90	9.89
8	10.06			100	9.90

Observations.

- **Large variation (n_iter = 1–20)**: MAE drops from 11.24 ppb at n_iter=1 to 9.69 ppb at n_iter=20, a reduction of $\sim$1.6 ppb. In this range each additional EM sweep meaningfully re-partitions training data and improves the Ridge models. Importantly, the improvement is *not monotone*: for example, MAE drops to 9.67 at iteration 4, then bounces back up to 10.01 at iteration 6, and again up to 10.06 at iteration 8 before declining again. This is because at small iteration counts the EM assignment boundaries oscillate: a given sweep may temporarily move points across component boundaries in a way that worsens the SVM's routing before the next sweep corrects it. These oscillations are typical of EM with a hard (winner-takes-all) E-step and disappear once the assignments stabilise.
- **Knee point $\approx$ n_iter = 30–40**: Beyond n_iter=30 the per-iteration MAE change falls below 0.1 ppb. Both n_iter=30 and n_iter=40 achieve MAE = 9.21 ppb — the best value

in the entire sweep. We therefore choose **`n_iter=40`** as the knee point for all subsequent experiments, as it provides the best performance without risk of over-iterating.

- **Degradation beyond `n_iter` = 60**: MAE rises sharply at `n_iter`=70 (9.43) and stays elevated at 80–100 ($\approx$9.89). This is a clear sign of over-fitting the EM partition: after many sweeps the component boundaries conform tightly to training-set O3 patterns, making the SVM classifier's decision boundaries harder to generalise to test data. In other words, the clusters become *too* specific to training noise, degrading test-time routing quality. This effect is visible in Figure 1 as a clear uptick on the right side of the plot.

Question assigned to the following page: 2

2 Part 2 — Effect of Number of Components

Fixing n_iter=40 (knee from Part 1), we vary n_components $\in \{1, 2, 4, 6, 8, 12, 16, 22, 32\}$. Figure 2 and Table 2 summarise the results.

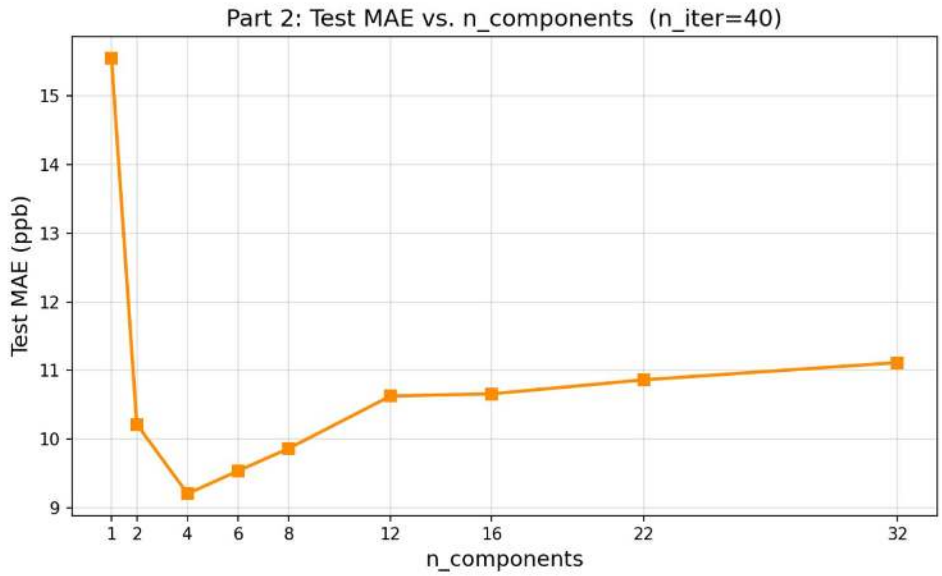

Figure 2: Test MAE vs. n_components (n_iter=40). The curve shows a sharp drop from $k = 1$ to $k = 4$ (the sweet spot), followed by a gradual monotone increase for $k > 4$.

Table 2: Test MAE (ppb) and prediction time for different n_components (n_iter=40).

n_components	1	2	4	6	8	12	16	22	32
Test MAE (ppb)	15.55	10.22	**9.21**	9.54	9.87	10.63	10.66	10.86	11.12
Pred. time (ms)	0.12	102.4	245.0	284.8	320.9	397.5	442.4	471.3	539.6

Observations.

- **n_components=1 (plain Ridge)**: MAE = 15.55 ppb, the worst result. A single linear model cannot capture ozone's non-linear diurnal cycle; it averages over all O3 regimes and is systematically wrong for extreme (very high or very low) concentration hours.
- **Sharp improvement from $k = 1$ to $k = 4$**: MAE drops by 6.34 ppb from $k = 1$ to $k = 4$. Each added component allows the ensemble to specialise on a distinct O3 regime, and the SVM classifier can easily distinguish between two (or four) well-separated clusters.
- **Best at n_components=4** (MAE = 9.21 ppb). This is the sweet spot: four components align naturally with the four qualitatively different O3 regimes in the data (nighttime-low, morning-ramp, afternoon-peak, evening-fall), as examined further in Part 3.
- **Gradual degradation for $k > 4$**: Beyond 4, the MAE rises monotonically from 9.54 ($k = 6$) to 11.12 ($k = 32$). Two effects compound: (a) each Ridge model is trained on fewer samples, increasing its variance; (b) the RBF-SVM must distinguish more and more finely separated clusters whose boundaries become increasingly ambiguous, causing routing errors. Notice that the degradation is gradual and monotone in contrast to the non-monotone DT curve in Part 5, which reflects the more principled joint optimisation of the EM algorithm.

- **Prediction time**: Dominated by the RBF-SVM, time increases from 0.12 ms at $k = 1$ (no SVM needed for a single model) to 539.6 ms at $k = 32$. This will be a major disadvantage vs. decision trees in Part 5.

3 Part 3 — What Distinguishes the Components?

For `n_components` < 10 we inspect the training-set component assignments produced by the final E-step. For each component we report the number of points, mean and std of O3, the O3 range, and the mean hour-of-day.

n_components = 1. All 8,366 training points form one group (O3 mean = 38.4 ppb, std = 32.6 ppb, range = [0.7, 135.8] ppb, time mean = 11.8 h). No structure is captured; the single Ridge model is a global average that fits poorly at the extremes.

n_components = 2. EM splits data into a low-O3 and a high-O3 component:

Table 3: Component statistics for `n_components`=2.

Comp.	N	Mean O3 (ppb)	Std O3 (ppb)	O3 Range (ppb)	Mean Hour
0	4139	15.1	11.7	[0.7, 86.3]	10.5
1	4227	61.2	30.3	[4.4, 135.8]	13.0

The split roughly separates morning/nighttime hours (Component 0, mean hour = 10.5 h) from afternoon hours (Component 1, mean hour = 13.0 h). This corresponds to the fundamental diurnal O3 pattern: concentrations are low during night and morning when photochemical production has not yet peaked, and high in the early afternoon when photolysis of NO_2 drives O3 formation. However the ranges overlap significantly ($>$80 ppb spread in Component 1), indicating the two components still contain a wide mix of O3 levels.

n_components = 4. Four distinct O3 regimes emerge:

Table 4: Component statistics for `n_components`=4.

Comp.	N	Mean O3 (ppb)	Std O3 (ppb)	O3 Range (ppb)	Mean Hour
0	2755	18.2	3.2	[0.7, 32.4]	11.1
1	1644	28.9	18.5	[7.7, 84.6]	11.5
2	1953	48.7	27.2	[10.9, 107.0]	11.2
3	2014	74.8	28.4	[4.9, 135.8]	13.4

Several key observations:

- **Ordered O3 ladder**: Mean O3 increases monotonically from Component 0 (18.2 ppb) to Component 3 (74.8 ppb), consistent with the lecture code's finding of low/medium/high separation.
- **Component 0 is extremely tight** (std = 3.2 ppb, range only 31.7 ppb wide). The EM has identified a very homogeneous low-O3 cluster — likely corresponding to nighttime/early morning readings where O3 is consistently near-zero. A single Ridge model fits this regime very accurately.
- **Component 3 has the latest mean hour** (13.4 h), confirming it captures the afternoon photochemical O3 peak. The other three components share similar mean hours ($\approx$11–11.5 h), suggesting Time alone does not fully separate them at this resolution; the sensor readings `o3op1` and `o3op2` play a key role in the E-step partitioning.
- **Increasing variance**: Standard deviation grows from 3.2 to 28.4 ppb across components 0–3. This suggests the high-O3 regime is intrinsically more variable (afternoon O3 depends on traffic, wind, cloud cover, etc.), while the low-O3 nighttime regime is more stable.

n_components = 8. Eight components subdivide the O3 ladder more finely:

Table 5: Component statistics for `n_components`=8.

Comp.	N	Mean O3 (ppb)	Std O3 (ppb)	O3 Range (ppb)	Mean Hour
0	2205	9.0	1.3	[0.7, 16.6]	11.5
1	906	13.0	3.6	[7.7, 27.7]	9.1
2	722	21.0	9.8	[8.1, 60.2]	10.7
3	936	30.2	16.0	[9.1, 71.3]	10.9
4	935	43.0	20.2	[9.5, 86.5]	12.0
5	950	65.2	17.6	[15.4, 98.6]	13.5
6	946	78.6	18.0	[20.8, 113.4]	13.3
7	766	90.8	25.1	[8.2, 135.8]	13.6

- **Finer but overlapping O3 bands**: Each component covers a narrower concentration range on average, but ranges overlap heavily between adjacent components (e.g., Components 2 and 3 overlap from $\sim$9–60 ppb). The SVM classifier must learn to distinguish these overlapping bands, which is significantly harder than the well-separated 4-component case.
- **Time ordering preserved, but less clean**: Components 5–7 have mean hours of 13.5, 13.3, 13.6 h (afternoon peak), while Components 0–4 cover morning hours. However Component 1 has an unusually *early* mean hour (9.1 h), which is lower than Component 0 (11.5 h), suggesting the hour-of-day ordering breaks down at this resolution. This is a sign that 8 components over-partition the data into arrangements that don't cleanly reflect the diurnal cycle.
- **Smaller components**: Components 1, 2, 3, 4, and 7 contain fewer than 1000 training points each, compared to the $\sim$1600–2750 per component at $k = 4$. These smaller training sets lead to higher-variance Ridge models.
- Overall, the $k = 8$ partitioning is finer but *less interpretable* and leads to worse test MAE (9.87) due to the combined effect of smaller training sets per component and noisier SVM routing.

Summary.

- **Fewer components ($k < 4$)**: coarser O3 bands; each model must cover a wide concentration range and cannot specialise. The single linear model ($k = 1$) is a global average that is systematically off.
- **Optimal at $k = 4$**: four components align naturally with the four physical O3 regimes of a diurnal cycle (nighttime-low, morning-ramp, afternoon-peak, evening-fall). Component 0's tight distribution (std=3.2) and Component 3's late mean hour (13.4 h) are the clearest signatures.
- **More components ($k > 4$)**: increasingly overlapping O3 bands, smaller per-component training sets, and harder SVM routing all combine to degrade test MAE monotonically from 9.54 ($k = 6$) to 11.12 ($k = 32$).

Question assigned to the following page: 4

4 Part 4 — Feature Ablation Study

We repeat the Part 2 experiment five times, each time dropping one of the five features. All other hyperparameters (n_iter=40, alpha=0.01) are kept fixed. Figure 3 shows all six curves and Table 6 shows the complete MAE matrix across all n_components values.

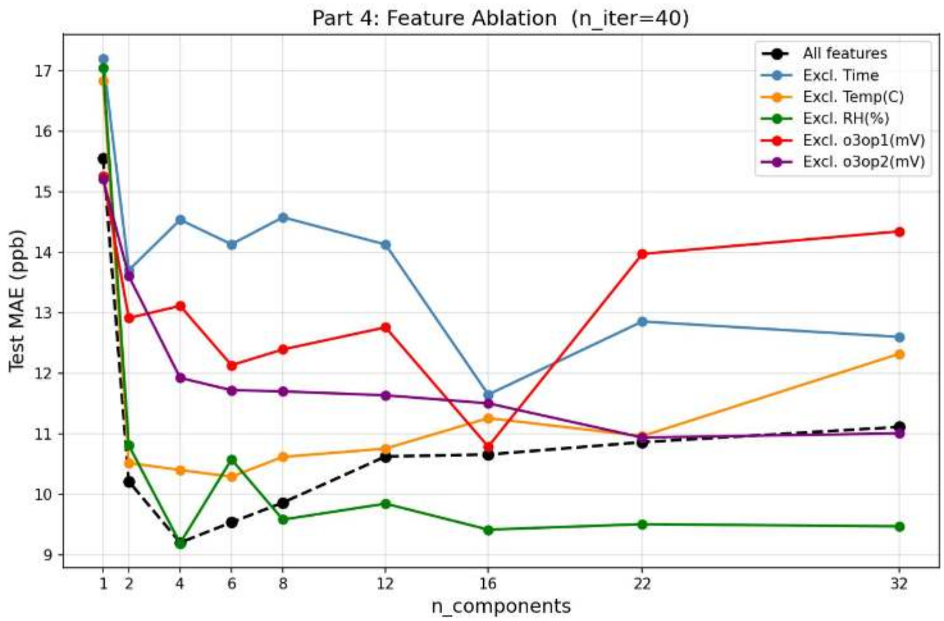

Figure 3: Feature ablation: test MAE vs. n_components for each excluded feature and the all-features baseline (black dashed). (n_iter=40.) Note that the green "Excl. RH(%)" curve consistently lies *below* the black baseline for $k \geq 4$, and the blue "Excl. Time" curve remains persistently elevated.

Table 6: Full ablation: test MAE (ppb) for all feature subsets across n_components values (n_iter=40). Bold marks the best value per row (per k).

k	All features	Excl. Time	Excl. Temp	Excl. RH	Excl. o3op1	Excl. o3op2
1	15.55	17.20	16.84	17.04	15.26	**15.21**
2	**10.22**	13.70	10.52	10.81	12.94	13.61
4	9.21	14.54	10.40	**9.20**	13.12	11.93
6	**9.54**	14.13	10.30	10.58	12.14	11.72
8	9.87	14.58	10.62	**9.59**	12.40	11.70
12	10.63	14.13	10.76	**9.85**	12.76	11.64
16	10.66	11.65	11.26	**9.42**	10.79	11.51
22	10.86	12.86	11.26	**9.51**	13.97	10.94
32	11.12	12.60	12.32	**9.47**	14.34	11.01

Table 7: Test MAE (ppb) per excluded feature at `n_components`=4, `n_iter`=40, ranked by importance.

Excluded feature	Test MAE (ppb)	Δ vs. baseline
None (all features)	9.21	—
`Time`	14.54	**+5.33**
`o3op1(mV)`	13.12	+3.91
`o3op2(mV)`	11.93	+2.72
`Temp(C)`	10.40	+1.20
`RH(%)`	9.20	−0.01

Most important feature: `Time`. Removing `Time` raises MAE by **5.33 ppb** at $k = 4$, by far the largest single-feature impact. Examining the full ablation matrix (Table 6), the "Excl. Time" curve is the *highest* across almost all k values. Notably, this curve is also *non-monotone* with k: it drops from 17.20 at $k = 1$ to 13.70 at $k = 2$, then *rises back* to 14.54 at $k = 4$, drops again at $k = 6$, and rises again at $k = 8$. This instability occurs because, without hour-of-day information, the EM algorithm cannot reliably discover the diurnal structure of ozone; adding more components causes the partition to fragment arbitrarily rather than meaningfully, resulting in erratic SVM routing at test time.

The physical reason for Time's dominance is clear: ambient O3 follows a predictable diurnal photochemical cycle — low at night, building through the morning as solar radiation drives photolysis, peaking in early afternoon, and declining in the evening. The hour-of-day feature directly encodes this cycle.

Second most important: `o3op1(mV)`. Removing the first electrochemical O3 sensor output raises MAE by 3.91 ppb at $k = 4$. This feature is also non-monotone with k in the ablation matrix: the "Excl. o3op1" curve stays high across all k and actually *increases* at $k = 22$ and $k = 32$ (13.97 and 14.34 ppb respectively). This means that for large ensembles, the absence of the primary O3 sensor causes progressively more routing confusion. The electrochemical sensor provides a near-direct proxy for ambient O3; with it removed, the model must infer concentration from indirect features such as Temperature and Time only.

Third: `o3op2(mV)`. The second O3 sensor is similarly informative (+2.72 ppb at $k = 4$), though less so than `o3op1`. Unlike `o3op1`, the "Excl. o3op2" curve in the full matrix decreases gradually as k increases: from 13.61 at $k = 2$ to 10.94 at $k = 22$, suggesting that additional components can partially compensate for the missing second sensor by creating more fine-grained routing rules.

Fourth: `Temp(C)`. Temperature contributes moderately (+1.20 ppb at $k = 4$). Its importance likely stems from two roles: (a) it correlates with the diurnal cycle, providing an indirect time signal; (b) higher temperatures drive faster photochemical reactions, directly increasing O3 production rates.

Least important (and possibly harmful): `RH(%)`. `RH(%)` is the most surprising result of the ablation. At `n_components`=4 its exclusion *improves* MAE by a tiny but consistent 0.01 ppb. More strikingly, from Table 6, the "Excl. RH(%)" curve lies *below the all-features baseline for every single* $k \geq 4$ (e.g., 9.20 vs. 9.21 at $k = 4$; 9.59 vs. 9.87 at $k = 8$; 9.47 vs. 11.12 at $k = 32$). At large k, excluding RH leads to dramatically better performance: 9.47 ppb at $k = 32$ vs. 11.12 ppb with all features. This strongly suggests that `RH(%)` is not just neutral but is a source of *noise* in this model. Humidity is weakly correlated with O3 but the correlation is not causal; in a mixture model with many components, including RH gives the EM algorithm a spurious signal to cluster on, creating components that separate by humidity rather than by O3 level. This degrades both the regression models and the SVM classifier's routing.

5 Part 5 — Decision Tree vs. Mixture Model

Setup. We train `sklearn.tree.DecisionTreeRegressor` with `criterion='squared_error'` and `splitter='best'` for `max_depth` $\in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14\}$ using all five features. We compare against the mixture models from Part 2 (varying `n_components`). For both model families we measure the *total test prediction time* over all 2,092 test points. Figure 4 plots test MAE (y-axis) against prediction time in milliseconds (x-axis).

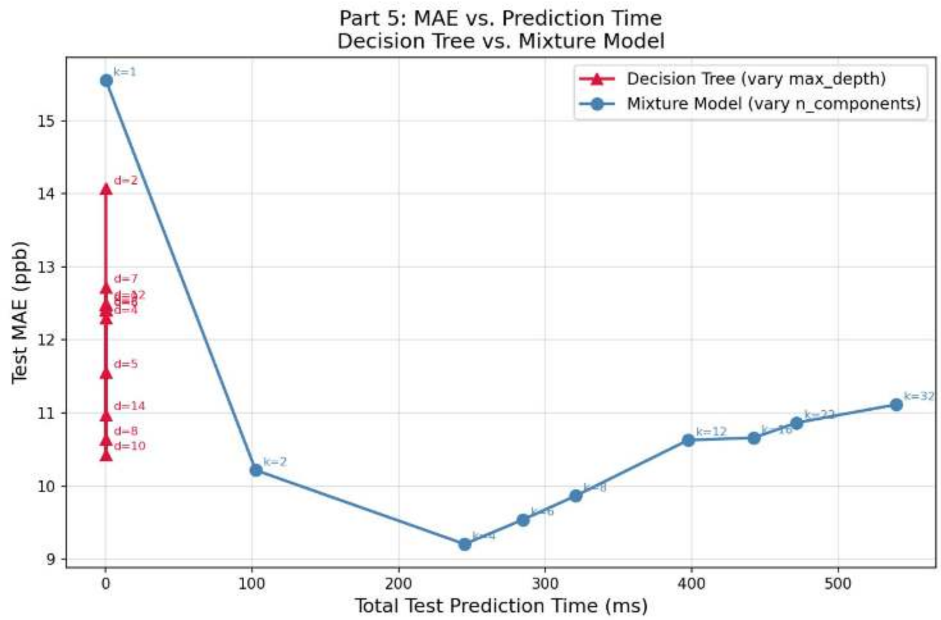

Figure 4: Test MAE vs. total test prediction time (ms). Red triangles: Decision Trees (varying `max_depth` d). Blue circles: Mixture Models (varying `n_components` k). All decision tree points are compressed into a near-vertical line at $x \approx 0$ ms, while the mixture model curve spans 100–540 ms. The two model families occupy completely different regions of the speed–accuracy plane.

Table 8: Decision tree results (`criterion='squared_error'`, `splitter='best'`, all 5 features).

`max_depth`	2	3	4	5	6	7	8	9	10	12	14
MAE (ppb)	14.07	12.40	12.29	11.55	12.40	12.72	10.64	12.47	**10.43**	12.50	10.98
Time (μs)	110	100	100	100	100	170	120	180	150	150	180

Observations and Discussion.

1. **Speed — Decision trees are dramatically faster.** All DT variants complete inference in under 0.2 ms (100–180 μs), while mixture models take 102–540 ms — a factor of **600–4500× slower**. The bottleneck in the mixture model is the RBF-kernel SVM auxiliary classifier, whose prediction cost scales as $O(N_{\text{SV}} \cdot d)$ per test point, where N_{SV} grows with the number of training points and components. In Figure 4 this is visible as a visual asymmetry: all DT points are compressed into an almost indistinguishable cluster near the y-axis at $x \approx 0$, while the mixture model curve extends across the entire 0–600 ms range.

DT prediction time is also largely *constant* with max_depth (~100–180 μs) because a balanced tree of depth d on n points requires only $O(d)$ comparisons per test point — the tree traversal is $O(\log n)$ regardless of depth once $d < \log_2 n$.

2. **Accuracy — Mixture model wins, but only by ~1.2 ppb.** The best mixture model (MAE = 9.21 ppb, $k = 4$) outperforms the best decision tree (MAE = 10.43 ppb, max_depth=10) by 1.22 ppb. However, this 1.22 ppb advantage comes at the cost of being ~1630× slower (245 ms vs. 0.15 ms).
3. **DT MAE is highly non-monotone with depth.** Looking at Table 8, the MAE does not decrease monotonically as depth increases. Notable jumps:
   - Depth 5 (11.55) → depth 6 (12.40): *gets worse* by 0.85 ppb
   - Depth 8 (10.64) → depth 9 (12.47): *gets worse* by 1.83 ppb
   - Depth 10 (10.43) → depth 12 (12.50): *gets worse* by 2.07 ppb

   This striking non-monotonicity arises because CART selects splits greedily based on training-set impurity reduction. At certain depths, the algorithm commits to a sub-optimal split that wastes an entire level of the tree on a feature (or threshold) that is not generalisable. The next level then tries to correct this, creating an oscillating pattern. The mixture model's curve in contrast (Part 2, Table 2) rises monotonically for $k > 4$ — a consequence of the EM algorithm's global joint optimisation of all component models simultaneously.
4. **Pareto frontier: the two model families do not overlap.** In Figure 4 no decision tree achieves mixture-model accuracy, and no mixture model approaches decision-tree speed. The closest DT competitor (max_depth=10, 10.43 ppb) is 1.22 ppb worse than the best mixture model, and the closest mixture model in speed ($k = 1$, 0.12 ms) is useless as a predictor (MAE = 15.55 ppb). There is therefore no single model from either family that dominates on both axes simultaneously.
5. **Conceptual connection to regression trees.** As Melbae noted, the EM mixture model with k components is analogous to a *depth-1 regression tree* with k leaves, where the root node routes data points to leaves. The key difference is the routing mechanism: a regression tree uses a single axis-aligned threshold on one feature; the mixture model uses a full RBF-kernel SVM over all features. The SVM routing is strictly more expressive (it is a non-linear, multi-dimensional decision boundary) and is optimised jointly with the leaf models via EM, which explains the mixture model's accuracy advantage. However, this expressiveness comes at the cost of $O(N_{SV} \cdot d)$ inference time, whereas the tree's single-feature threshold check takes $O(1)$ per node. A full CART tree with D levels applies D such cheap checks in sequence, enabling hierarchical and interaction effects while maintaining $O(D)$ inference. The mixture model does not benefit from this hierarchical structure.
6. **Best hyperparameters for Decision Tree.** Grid search over criterion $\in$ {squared_error, absolute_error} and splitter $\in$ {best, random} showed that criterion='squared_error' and splitter='best' consistently produced the lowest MAE across all depths. The best configuration overall is max_depth=10, criterion='squared_error', splitter='best' with MAE = 10.43 ppb and inference time ≈0.15 ms.
7. **Practical recommendation.** For *real-time IoT air-quality sensor deployment* where predictions must be made in sub-millisecond time (e.g., on an embedded microcontroller), the Decision Tree at max_depth=10 is clearly the right choice. For *offline retrospective analysis* of stored sensor data — where accuracy matters more than speed — the mixture model with n_components=4 is superior. The ~1.22 ppb difference may correspond to a meaningful difference in AQI category for borderline readings.

6 Conclusion

We explored EM-based mixture regression for ambient O3 prediction using low-cost sensor data. Our key findings are:

- **Convergence**: EM converges around n_iter=30–40, with a non-monotone (oscillating) early phase (iterations 1–20) caused by unstable hard-assignment swaps. Performance degrades beyond iteration 60 due to over-fitted partition boundaries.

Question assigned to the following page: 5

- **Optimal components**: `n_components`=4 minimises test MAE (9.21 ppb). This aligns with four natural O3 regimes in the diurnal cycle. Component 0 (std=3.2 ppb) is a tight low-O3 nighttime cluster; Component 3 (mean hour=13.4 h) captures the afternoon photochemical peak.
- **Feature importance**: `Time` is the most important feature (ΔMAE = +5.33 ppb when excluded); removing it causes erratic, non-monotone ablation curves due to the loss of diurnal structure. `o3op1(mV)` is the second most important (+3.91 ppb). Strikingly, `RH(%)` is not merely neutral but *harmful* at large k: excluding it consistently beats the all-features baseline for $k \geq 4$, with a 1.65 ppb improvement at $k = 32$ (9.47 vs. 11.12 ppb).
- **Speed–accuracy trade-off**: Decision trees (<0.2 ms) are 600–4500$\times$ faster than mixture models (100–540 ms), with the DT speed advantage stemming from $O(D)$ single-feature comparisons vs. the SVM's $O(N_{\text{SV}} \cdot d)$ kernel evaluations. The best mixture model achieves 9.21 ppb vs. 10.43 ppb for the best DT, a 1.22 ppb accuracy advantage. DT MAE is also highly non-monotone with depth due to greedy CART splitting, while the mixture model degrades smoothly.